

Module 4

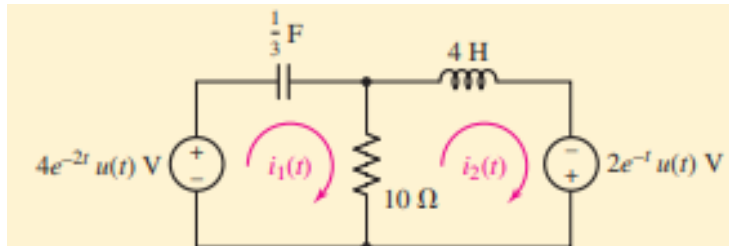
Circuit Analysis in the S Domain

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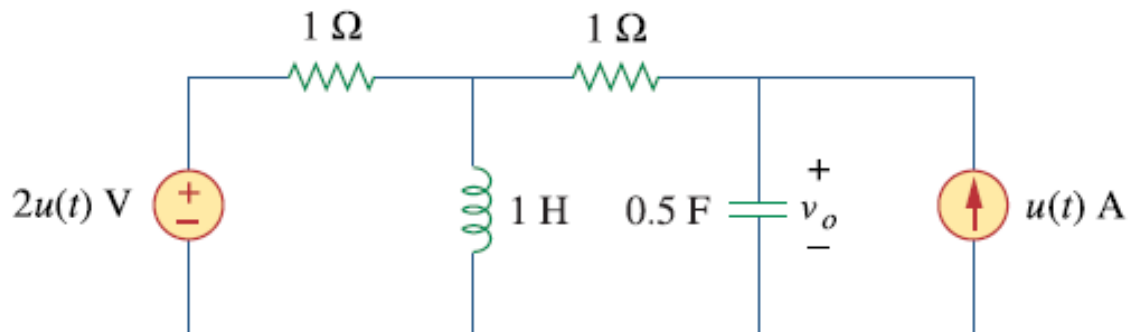
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Tutorial

1. Determine the two mesh currents i_1 and i_2 in the circuit below. There is no energy initially stored in the circuit.

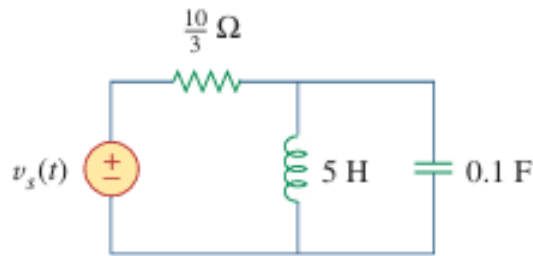


2. Find $v_o(t)$ for all time greater than zero in the circuit given below.



Tutorial

3. Consider the circuit below. Find the value of the voltage across the capacitor assuming that the value of $v_s(t) = 10 u(t)$ V and assume that at $t = 0$, -1 A flows through the inductor and +5 V is across the capacitor.



4. For the circuit in problem 3, find the voltage across the capacitor applying superposition theorem assuming the same initial conditions.

5. For the circuit in problem 3, find the current through the inductor for $t > 0$ assuming the same initial conditions.